

Customer Churn Analysis for Telecom Industry

Elevate Labs – Data Analyst Internship

Abstract

This project analyzes the IBM Telco Customer Churn dataset to understand customer behavior, identify factors influencing churn, and provide business insights. Python, SQL, and machine learning techniques were used to support data-driven decision making.

Objectives

- Clean and preprocess telecom customer data.
- Perform exploratory data analysis.
- Build and evaluate machine learning models.
- Explain predictions using SHAP.
- Perform SQL-based business analysis.
- Segment customers and provide recommendations.

Dataset

The dataset contains 7,043 customer records with 21 features, including demographic details, service subscriptions, contract type, payment method, monthly charges, tenure, and churn status.

Methodology

The workflow included data cleaning, missing value handling, feature encoding, exploratory data analysis, model training using Logistic Regression, Decision Tree, and Random Forest, SHAP explainability, SQL analysis, and customer segmentation.

Results

The Random Forest model achieved the best predictive performance among the evaluated models. SQL analysis identified customer distribution, churn rate, contract-wise trends, and payment behavior. Customer segmentation classified customers into Loyal, At Risk, and Dormant groups.

Business Recommendations

- Encourage long-term contracts.
- Improve technical support services.
- Target at-risk customers with personalized offers.
- Reward loyal customers.
- Monitor churn using predictive analytics.

Conclusion

The project demonstrates how data analytics and machine learning can help telecom companies understand customer behavior, reduce churn, and improve retention strategies.